

Reference: random full-wave search
200 calls to reach regret 0.10, against **1 call** with a surrogate

Pool of designs
1000 random discs
4 parameters each

Surrogate screens all
FNO or encoder
cheap; rank by
figure of merit

Verify the top M
full-wave solve
 $M = 1$ or 5

Best verified design
regret 0.064 (FNO),
0.013 (encoder), $M = 1$

Certificate gating (variant)
top 50: accept if
 $\text{bound} \leq \tau = 0.2$,
else full-wave

28 to 41 full-wave calls on average, without improving the design